$\emptyset : \mathbb{R}^n \to \mathbb{R}^{n'}$ $\emptyset$ = Fhi

$x \in \mathbb{R}$ $x' = \emptyset(x) = (x, x^2, x^3, x^4)$

$x' = \emptyset(x) = (\emptyset_1(x), \emptyset_2(x), \cdots \emptyset_{n'}(x))$

Expansion polinomial

$x = (x_1, x_2)$

$\emptyset = (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$

$\emptyset = (x_1, x_2, x_1^2, x_1 x_2, x_2^2\ x^3, x_1^2 x_2, x_1 x_2^2, x_2^3)$

$x = (x_1, x_2, x_3, x_4)$

$\emptyset = (x_1, x_2, x_3\ x_4, x_1^2, x_1 x_2, x_1 x_3, x_1 x_4, x_2^2, x_2 x_3, x_2 x_4, x_3^2, x_3 x_4, x_4^2)$

$H = \{h\}$ tal que

$h(x) = w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + w_4 x_1^4 + w_5 x_1^5 + w_6 x_1^6 + w_7 x_1^7 + w_8 x_1^8 + b$

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$h(x) = w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + w_4 x_1^4 + w_5 x_1^5 + w_6 x_1^6 + w_7 x_1^7 + w_8 x_1^8 + b$

bajo $w_j = 0$ si $j \geq 3$

$h(x) = w_1 x_1 + w_2 x_2 + b$

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Dominio c = cantidad de variables libres

$h(x) = w^T \emptyset(x) + b$ $w, \emptyset(x) \in \mathbb{R}^{n'}, b \in \mathbb{R}$

n'

$h(x) = w^T \phi(x) + b \quad w, \phi(x) \in \mathbb{R}^{n'}, \; b \in \mathbb{R}$

$$\text{bajo} \; \underbrace{\sum_{j=1}^{n'} w_j^2 \leq C}_{w^T w \leq C} \leftarrow \text{Regularización}$$

Regularización blanda

$\|w\|_2 = \sqrt{w^T w}$

$\|w\|_1 = \sum_{j=1}^{n'} |w_j|$